Proof We begin with the hypothesis that f is a rational function, that is, $f(z) = p(z)/q(z)$, where p and q are polynomials. We know that f is analytic for all z except at the zeros of q. Suppose z_0 is a zero of q but not of p. Then Theorem 6.4.1 tell us that there exists a positive integer n such that $q(z) = (z - z_0)^n Q(z)$, where Q is a polynomial and $Q(z_0) \neq 0$. Now invoke Theorem 6.4.2. Consider one more case to finish the proof.

6.5 Residues and Residue Theorem

INTRODUCTION We saw in the last section that if a complex function f has an *isolated singularity* at a point z_0, then f has a Laurent series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots,$$

which converges for all z *near* z_0. More precisely, the representation is valid in some deleted neighborhood of z_0 or punctured open disk $0 < |z - z_0| < R$. In this section our entire focus will be on the coefficient a_{-1} and its importance in the evaluation of contour integrals.

Residue The coefficient a_{-1} of $1/(z - z_0)$ in the preceding Laurent series is called the **residue** of the function f at the isolated singularity z_0. We shall use the notation

$$a_{-1} = \text{Res}(f(z),\ z_0)$$

to denote the residue of f at z_0. Recall, if the principal part of the Laurent series valid for $0 < |z - z_0| < R$ contains a finite number of terms with a_{-n} the last nonzero coefficient, then z_0 is a pole of order n; if the principal part of the series contains an infinite number of terms with nonzero coefficients, then z_0 is an essential singularity.

EXAMPLE 1 Residues

(**a**) In part (b) of Example 2 in Section 6.4 we saw that $z = 1$ is a pole of order two of the function $f(z) = \dfrac{1}{(z-1)^2(z-3)}$. From the Laurent series obtained in that example valid for the deleted neighborhood of $z = 1$ defined by $0 < |z - 1| < 2$,

$$f(z) = \frac{-1/2}{(z-1)^2} + \overbrace{\frac{-1/4}{z-1}}^{a_{-1}} - \frac{1}{8} - \frac{z-1}{16} - \cdots$$

we see that the coefficient of $1/(z - 1)$ is $a_{-1} = \text{Res}(f(z), 1) = -\frac{1}{4}$.

(**b**) In Example 6 of Section 6.3 we saw that $z = 0$ is an essential singularity of $f(z) = e^{3/z}$. Inspection of the Laurent series obtained in that example,

$$e^{3/z} = 1 + \overbrace{\frac{3}{z}}^{a_{-1}} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots,$$

$0 < |z| < \infty$, shows that the coefficient of $1/z$ is $a_{-1} = \text{Res}(f(z),\ 0) = 3$. ☰

We will see why the coefficient a_{-1} is so important later on in this section. In the meantime we are going to examine ways of obtaining this complex number when z_0 is a *pole* of a function f without the necessity of expanding f in a Laurent series at z_0. We begin with the residue at a simple pole.

THEOREM 6.5.1 Residue at a Simple Pole

If f has a simple pole at $z = z_0$, then

$$\text{Res}(f(z), z_0) = \lim_{z \to z_0} (z - z_0) f(z). \tag{1}$$

Proof Since f has a simple pole at $z = z_0$, its Laurent expansion convergent on a punctured disk $0 < |z - z_0| < R$ has the form

$$f(z) = \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0) + \cdots,$$

where $a_{-1} \neq 0$. By multiplying both sides of this series by $z - z_0$ and then taking the limit as $z \to z_0$ we obtain

$$\begin{aligned} \lim_{z \to z_0} (z - z_0) f(z) &= \lim_{z \to z_0} [a_{-1} + a_0(z - z_0) + a_1(z - z_0)^2 + \cdots] \\ &= a_{-1} = \text{Res}(f(z), z_0). \end{aligned}$$

THEOREM 6.5.2 Residue at a Pole of Order *n*

If f has a pole of order n at $z = z_0$, then

$$\text{Res}(f(z), z_0) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} \Big[(z - z_0)^n f(z) \Big]. \tag{2}$$

Proof Because f is assumed to have pole of order n at $z = z_0$, its Laurent expansion convergent on a punctured disk $0 < |z - z_0| < R$ must have the form

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots,$$

where $a_{-n} \neq 0$. We multiply the last expression by $(z - z_0)^n$,

$$(z - z_0)^n f(z) = a_{-n} + \cdots + a_{-2}(z - z_0)^{n-2} + a_{-1}(z - z_0)^{n-1} + a_0(z - z_0)^n + a_1(z - z_0)^{n+1} + \cdots$$

and then differentiate both sides of the equality $n - 1$ times:

$$\frac{d^{n-1}}{dz^{n-1}} \Big[(z - z_0)^n f(z) \Big] = (n-1)! a_{-1} + n! a_0(z - z_0) + \cdots. \tag{3}$$

Since all the terms on the right-hand side after the first involve positive integer powers of $z - z_0$, the limit of (3) as $z \to z_0$ is

$$\lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} \Big[(z - z_0)^n f(z) \Big] = (n-1)! a_{-1}.$$

Solving the last equation for a_{-1} gives (2).

Notice that (2) reduces to (1) when $n = 1$.

EXAMPLE 2 Residue at a Pole

The function $f(z) = \dfrac{1}{(z-1)^2(z-3)}$ has a simple pole at $z = 3$ and a pole of order 2 at $z = 1$. Use Theorems 6.5.1 and 6.5.2 to find the residues.

Solution Since $z = 3$ is a simple pole, we use (1):

$$\text{Res}(f(z), 3) = \lim_{z \to 3} (z - 3) f(z) = \lim_{z \to 3} \frac{1}{(z-1)^2} = \frac{1}{4}.$$

Now at the pole of order 2, the result in (2) gives

$$\begin{aligned}\text{Res}(f(z), 1) &= \frac{1}{1!}\lim_{z\to 1}\frac{d}{dz}\left[(z-1)^2 f(z)\right] \\ &= \lim_{z\to 1}\frac{d}{dz}\frac{1}{z-3} \\ &= \lim_{z\to 1}\frac{-1}{(z-3)^2} = -\frac{1}{4}.\end{aligned}$$

When f is not a rational function, calculating residues by means of (1) or (2) can sometimes be tedious. It is possible to devise alternative residue formulas. In particular, suppose a function f can be written as a quotient $f(z) = g(z)/h(z)$, where g and h are analytic at $z = z_0$. If $g(z_0) \neq 0$ and if the function h has a zero of order 1 at z_0, then f has a simple pole at $z = z_0$ and

An alternative method for computing a residue at a simple pole ▶

$$\text{Res}(f(z), z_0) = \frac{g(z_0)}{h'(z_0)}. \tag{4}$$

To derive this result we shall use the definition of a zero of order 1, the definition of a derivative, and then (1). First, since the function h has a zero of order 1 at z_0, we must have $h(z_0) = 0$ and $h'(z_0) \neq 0$. Second, by definition of the derivative given in (12) of Section 3.2,

$$h'(z_0) = \lim_{z\to z_0}\frac{h(z) - \overbrace{h(z_0)}^{0}}{z - z_0} = \lim_{z\to z_0}\frac{h(z)}{z - z_0}.$$

We then combine the preceding two facts in the following manner in (1):

$$\text{Res}(f(z), z_0) = \lim_{z\to z_0}(z - z_0)\frac{g(z)}{h(z)} = \lim_{z\to z_0}\frac{g(z)}{\dfrac{h(z)}{z - z_0}} = \frac{g(z_0)}{h'(z_0)}.$$

There are several alternative ways of arriving at formula (4). For instance, it can be obtained by a single application of L'Hôpital's rule (page 126), but you are asked in Problem 42 in Exercises 6.5 to derive (4) using (5) of Section 6.4. Residue formulas for poles of order greater than 1 are far more complicated than (4) and will not be presented here. Practicality aside, a derivation of one of these higher-order formulas provides an opportunity to review and use important concepts. See Problem 43 in Exercises 6.5.

EXAMPLE 3 Using (4) to Compute Residues

The polynomial $z^4 + 1$ can be factored as $(z - z_1)(z - z_2)(z - z_3)(z - z_4)$, where z_1, z_2, z_3, and z_4 are the four distinct roots of the equation $z^4 + 1 = 0$ (or equivalently, the four fourth roots of -1). It follows from Theorem 6.4.3 that the function

$$f(z) = \frac{1}{z^4 + 1}$$

has four simple poles. Now from (4) of Section 1.4 we have $z_1 = e^{\pi i/4}$, $z_2 = e^{3\pi i/4}$, $z_3 = e^{5\pi i/4}$, and $z_4 = e^{7\pi i/4}$. To compute the residues, we use (4) of this

section along with Euler's formula (6) of Section 1.6:

$$\text{Res}(f(z), z_1) = \frac{1}{4z_1^3} = \frac{1}{4}e^{-3\pi i/4} = -\frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i$$

$$\text{Res}(f(z), z_2) = \frac{1}{4z_2^3} = \frac{1}{4}e^{-9\pi i/4} = \frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i$$

$$\text{Res}(f(z), z_3) = \frac{1}{4z_3^3} = \frac{1}{4}e^{-15\pi i/4} = \frac{1}{4\sqrt{2}} + \frac{1}{4\sqrt{2}}i$$

$$\text{Res}(f(z), z_4) = \frac{1}{4z_4^3} = \frac{1}{4}e^{-21\pi i/4} = -\frac{1}{4\sqrt{2}} + \frac{1}{4\sqrt{2}}i.$$

Of course, we could have calculated each of the residues in Example 3 using formula (1). But the procedure in this case would have entailed substantially more algebra. For example, we first use the factorization of $z^4 + 1$ to write f as:

$$f(z) = \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)}.$$

By (1) the residue at, say, the pole z_1 is

$$\text{Res}(f(z), z_1) = \lim_{z \to z_1} (z - z_1) \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)}$$

$$= \frac{1}{(z_1 - z_2)(z_1 - z_3)(z_1 - z_4)}$$

$$= \frac{1}{(e^{\pi i/4} - e^{3\pi i/4})(e^{\pi i/4} - e^{5\pi i/4})(e^{\pi i/4} - e^{7\pi i/4})}.$$

Then we face the daunting task of simplifying the denominator of the last expression. Finally, we must do this process three more times.

Residue Theorem We come now to the reason why the residue concept is important. The next theorem states that under some circumstances we can evaluate complex integrals $\oint_C f(z)\,dz$ by summing the residues at the isolated singularities of f within the closed contour C.

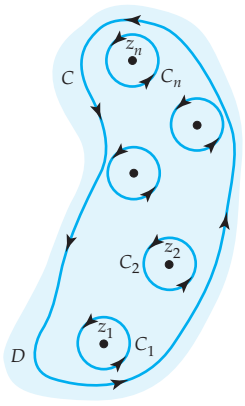


FIGURE 6.5.1 n singular points within contour C

THEOREM 6.5.3 Cauchy's Residue Theorem

Let D be a simply connected domain and C a simple closed contour lying entirely within D. If a function f is analytic on and within C, except at a finite number of isolated singularities $z_1, z_2, \ldots, z_n$ within C, then

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \text{Res}\,(f(z), z_k). \tag{5}$$

Proof Suppose $C_1, C_2, \ldots, C_n$ are circles centered at $z_1, z_2, \ldots, z_n$, respectively. Suppose further that each circle C_k has a radius r_k small enough so that $C_1, C_2, \ldots, C_n$ are mutually disjoint and are interior to the simple closed curve C. See FIGURE 6.5.1. Now in (20) of Section 6.3 we saw that $\oint_{C_k} f(z)\,dz = 2\pi i\,\text{Res}(f(z), z_k)$, and

so by Theorem 5.3.2 we have

$$\oint_C f(z)\,dz = \sum_{k=1}^{n} \oint_{C_k} f(z)\,dz = 2\pi i \sum_{k=1}^{n} \text{Res}\,(f(z), z_k).$$

EXAMPLE 4 Evaluation by the Residue Theorem

Evaluate $\displaystyle\oint_C \frac{1}{(z-1)^2(z-3)}\,dz$, where

(**a**) the contour C is the rectangle defined by $x = 0$, $x = 4$, $y = -1$, $y = 1$,

(**b**) and the contour C is the circle $|z| = 2$.

Solution (**a**) Since both $z = 1$ and $z = 3$ are poles within the rectangle we have from (5) that

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\,[\text{Res}(f(z), 1) + \text{Res}(f(z), 3)]$$

We found these residues in Example 2. Therefore,

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\left[\left(-\frac{1}{4}\right) + \frac{1}{4}\right] = 0.$$

(**b**) Since only the pole $z = 1$ lies within the circle $|z| = 2$, we have from (5)

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\,\text{Res}(f(z), 1) = 2\pi i\left(-\frac{1}{4}\right) = -\frac{\pi}{2}i.$$

EXAMPLE 5 Evaluation by the Residue Theorem

Evaluate $\displaystyle\oint_C \frac{2z+6}{z^2+4}\,dz$, where the contour C is the circle $|z - i| = 2$.

Solution By factoring the denominator as $z^2 + 4 = (z - 2i)(z + 2i)$ we see that the integrand has simple poles at $-2i$ and $2i$. Because only $2i$ lies within the contour C, it follows from (5) that

$$\oint_C \frac{2z+6}{z^2+4}\,dz = 2\pi i\,\text{Res}(f(z), 2i).$$

But

$$\text{Res}(f(z), 2i) = \lim_{z\to 2i}(z - 2i)\frac{2z+6}{(z-2i)(z+2i)} = \frac{6+4i}{4i} = \frac{3+2i}{2i}.$$

Hence,

$$\oint_C \frac{2z+6}{z^2+4}\,dz = 2\pi i\left(\frac{3+2i}{2i}\right) = \pi(3+2i).$$

EXAMPLE 6 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{e^z}{z^4+5z^3}\,dz$, where the contour C is the circle $|z| = 2$.

Solution Writing the denominator as $z^4 + 5z^3 = z^3(z + 5)$ reveals that the integrand $f(z)$ has a pole of order 3 at $z = 0$ and a simple pole at $z = -5$. But only the pole $z = 0$ lies within the given contour and so from (5) and (2) we have,

$$\oint_C \frac{e^z}{z^4+5z^3}\,dz = 2\pi i\,\text{Res}(f(z), 0) = 2\pi i\,\frac{1}{2!}\lim_{z\to 0}\frac{d^2}{dz^2}\left[z^3\cdot\frac{e^z}{z^3(z+5)}\right]$$
$$= \pi i\lim_{z\to 0}\frac{(z^2+8z+17)e^z}{(z+5)^3} = \frac{17\pi}{125}i.$$

EXAMPLE 7 Evaluation by the Residue Theorem

Evaluate $\oint_C \tan z\,dz$, where the contour C is the circle $|z| = 2$.

Solution The integrand $f(z) = \tan z = \sin z/\cos z$ has simple poles at the points where $\cos z = 0$. We saw in (21) of Section 4.3 that the only zeros of $\cos z$ are the real numbers $z = (2n + 1)\pi/2$, $n = 0, \pm 1, \pm 2, \ldots$. Since only $-\pi/2$ and $\pi/2$ are within the circle $|z| = 2$, we have

$$\oint_C \tan z\,dz = 2\pi i\left[\text{Res}\left(f(z), -\frac{\pi}{2}\right) + \text{Res}\left(f(z), \frac{\pi}{2}\right)\right].$$

With the identifications $g(z) = \sin z$, $h(z) = \cos z$, and $h'(z) = -\sin z$, we see from (4) that

$$\text{Res}\left(f(z), -\frac{\pi}{2}\right) = \frac{\sin(-\pi/2)}{-\sin(-\pi/2)} = -1$$

and

$$\text{Res}\left(f(z), \frac{\pi}{2}\right) = \frac{\sin(\pi/2)}{-\sin(\pi/2)} = -1.$$

Therefore,

$$\oint_C \tan z\,dz = 2\pi i[-1-1] = -4\pi i.$$

Theorem 6.5.3 is applicable at an essential singularity.

▶ Although there is no nice formula analogous to (1), (2), or (4), for computing the residue at an essential singularity z_0, the next example shows that Theorem 6.5.3 is still applicable at z_0.

EXAMPLE 8 Evaluation by the Residue Theorem

Evaluate $\oint_C e^{3/z}dz$, where the contour C is the circle $|z| = 1$.

Solution As we have seen, $z = 0$ is an essential singularity of the integrand $f(z) = e^{3/z}$ and so neither formulas (1) and (2) are applicable to find the residue of f at that point. Nevertheless, we saw in Example 1 that the Laurent series of f at $z = 0$ gives $\text{Res}(f(z),\ 0) = 3$. Hence from (5) we have

$$\oint_C e^{3/z}\,dz = 2\pi i\,\text{Res}(f(z), 0) = 2\pi i\,(3) = 6\pi i.$$

EXERCISES 6.5

Answers to selected odd-numbered problems begin on page ANS-18.

In Problems 1–6, use an appropriate Laurent series to find the indicated residue.

1. $f(z) = \dfrac{2}{(z-1)(z+4)}$; $\text{Res}(f(z), 1)$

2. $f(z) = \dfrac{1}{z^3(1-z)^3}$; $\text{Res}(f(z), 0)$

3. $f(z) = \dfrac{4z-6}{z(2-z)}$; $\text{Res}(f(z), 0)$

4. $f(z) = (z+3)^2 \sin\left(\dfrac{2}{z+3}\right)$; $\text{Res}(f(z), -3)$

5. $f(z) = e^{-2/z^2}$; $\text{Res}(f(z), 0)$

6. $f(z) = \dfrac{e^{-z}}{(z-2)^2}$; $\text{Res}(f(z), 2)$

In Problems 7–16, use (1), (2), or (4) to find the residue at each pole of the given function.

7. $f(z) = \dfrac{z}{z^2+16}$

8. $f(z) = \dfrac{4z+8}{2z-1}$

9. $f(z) = \dfrac{1}{z^4+z^3-2z^2}$

10. $f(z) = \dfrac{1}{(z^2-2z+2)^2}$

11. $f(z) = \dfrac{5z^2-4z+3}{(z+1)(z+2)(z+3)}$

12. $f(z) = \dfrac{2z-1}{(z-1)^4(z+3)}$

13. $f(z) = \dfrac{\cos z}{z^2(z-\pi)^3}$

14. $f(z) = \dfrac{e^z}{e^z-1}$

15. $f(z) = \sec z$

16. $f(z) = \dfrac{1}{z \sin z}$

In Problems 17–20, use Cauchy's residue theorem, where appropriate, to evaluate the given integral along the indicated contours.

17. $\displaystyle\oint_C \frac{1}{(z-1)(z+2)^2}\,dz$ **(a)** $|z| = \frac{1}{2}$ **(b)** $|z| = \frac{3}{2}$ **(c)** $|z| = 3$

18. $\displaystyle\oint_C \frac{z+1}{z^2(z-2i)}\,dz$ **(a)** $|z| = 1$ **(b)** $|z-2i| = 1$ **(c)** $|z-2i| = 4$

19. $\displaystyle\oint_C z^3 e^{-1/z^2}\,dz$ **(a)** $|z| = 5$ **(b)** $|z+i| = 2$ **(c)** $|z-3| = 1$

20. $\displaystyle\oint_C \frac{1}{z \sin z}\,dz$ **(a)** $|z-2i| = 1$ **(b)** $|z-2i| = 3$ **(c)** $|z| = 5$

In Problems 21–34, use Cauchy's residue theorem to evaluate the given integral along the indicated contour.

21. $\displaystyle\oint_C \frac{1}{z^2+4z+13}\,dz$, C: $|z-3i| = 3$

22. $\displaystyle\oint_C \frac{1}{z^3(z-1)^4}\,dz$, C: $|z-2| = \frac{3}{2}$

23. $\displaystyle\oint_C \frac{z}{z^4-1}\,dz$, C: $|z| = 2$

24. $\displaystyle\oint_C \frac{z}{(z+1)(z^2+1)}\,dz$, C: $16x^2 + y^2 = 4$

25. $\displaystyle\oint_C \frac{ze^z}{z^2-1}\,dz$, C: $|z| = 2$

26. $\displaystyle\oint_C \frac{e^z}{z^3+2z^2}\,dz$, C: $|z|=3$

27. $\displaystyle\oint_C \frac{\tan z}{z}\,dz$, C: $|z-1|=2$

28. $\displaystyle\oint_C \frac{\cot \pi z}{z^2}\,dz$, C: $|z|=\frac{1}{2}$

29. $\displaystyle\oint_C \cot \pi z\,dz$, C is the rectangle defined by $x=\frac{1}{2}$, $x=\pi$, $y=-1$, $y=1$

30. $\displaystyle\oint_C \frac{2z-1}{z^2(z^3+1)}\,dz$, C is the rectangle defined by $x=-2$, $x=1$, $y=-\frac{1}{2}$, $y=1$

31. $\displaystyle\oint_C \left(z^2 e^{1/(\pi z)}+\frac{ze^z}{z^4-\pi^4}\right)dz$, C: $4x^2+y^2=16$

32. $\displaystyle\oint_C \frac{\cos z}{(z-1)^2(z^2+9)}\,dz$, C: $|z-1|=1$

33. $\displaystyle\oint_C \frac{1}{z^6+1}\,dz$, C is the semicircle defined by $y=0$, $y=\sqrt{4-x^2}$

34. $\displaystyle\oint_C e^{4/(z-2)}\,dz$, C: $|z-1|=3$

Focus on Concepts

35. (**a**) Use series to show that $z=0$ is a zero of order 2 of $1-\cos z$.

(**b**) In view of part (a), $z=0$ is a pole of order two of the function $f(z)=e^z/(1-\cos z)$ and hence has a Laurent series

$$f(z)=\frac{e^z}{1-\cos z}=\frac{a_{-2}}{z^2}+\frac{a_{-1}}{z}+a_0+a_1z+a_2z^2+\cdots$$

valid for $0<|z|<2\pi$. Use series for e^z and $1-\cos z$ and equate coefficients in the product

$$e^z=(1-\cos\,z)\left(\frac{a_{-2}}{z^2}+\frac{a_{-1}}{z}+a_0+\cdots\right)$$

to determine a_{-2}, a_{-1}, and a_0.

(**c**) Evaluate $\displaystyle\oint_C \frac{e^z}{1-\cos z}\,dz$, where C is $|z|=1$.

36. Discuss how to evaluate $\displaystyle\oint_C e^{1/z}\sin\left(\frac{1}{z}\right)dz$, where C is $|z|=1$. Carry out your ideas.

37. Consider the function $f(z)=z^4/(1-z^{1/2})$, where $z^{1/2}$ denotes the principal branch of the square root function. Discuss and justify your answer: Does f have a pole at $z=1$? If so, find Res$(f(z),\ 1)$.

38. Residues can be used to find coefficients in partial fraction decompositions of rational functions. Suppose that $p(z)$ is a polynomial of degree $< n$ and that $z_1, z_2, \ldots, z_n$ are n distinct complex numbers that are not zeros of $p(z)$. Then

$$f(z)=\frac{p(z)}{(z-z_1)(z-z_2)\cdots(z-z_n)}=\frac{A_1}{z-z_1}+\frac{A_2}{z-z_2}+\cdots+\frac{A_n}{z-z_n}$$

where $A_1, A_2, \ldots, A_n$ are complex constants.

(**a**) Use (1) to show

$$\text{Res}(f(z), z_j)=\frac{p(z_j)}{(z_j-z_1)\cdots(z_j-z_{j-1})(z_j-z_{j+1})\cdots(z_j-z_n)}.$$

(**b**) Now use a Laurent series to prove that $A_j = \text{Res}(f(z), z_j)$. [*Hint*: All of the terms $A_k/(z - z_k)$ where $k \neq j$ are analytic at z_j.]

39. Use Problem 38 to find the partial fraction decomposition of $f(z) = \dfrac{5z^2 - z + 2}{z(z+1)(z-i)}$.

40. The partial fraction method in Problem 38 is only valid when the denominator of the rational function $f(z)$ is a product of distinct linear factors. Residues can also be used, however, to deal with factors of higher order. For example, suppose that $p(z)$ is a polynomial of degree ≤ 2 and that z_1 and z_2 are distinct complex numbers that are not zeros of $p(z)$. Then

$$f(z) = \frac{p(z)}{(z - z_1)(z - z_2)^2} = \frac{A}{z - z_1} + \frac{B}{z - z_2} + \frac{C}{(z - z_2)^2}$$

where A, B, and C are complex constants. Use Laurent series and residues to show

$$A = \frac{p(z_1)}{(z_1 - z_2)^2} \qquad B = \frac{d}{dz}\left[\frac{p(z)}{z - z_1}\right]_{z=z_2} \qquad C = \frac{p(z_2)}{z_2 - z_1}.$$

41. Use Problem 40 to find the partial fraction decomposition of $f(z) = \dfrac{5z^2 - z + 2}{z(z-i)^2}$.

42. If $h(z)$ has a zero of order 1, it follows from (5) of Section 6.4 that $h(z) = (z - z_0)\,\phi(z)$, where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$. With $f(z) = g(z)/h(z)$, use (1) to derive (4):

$$\text{Res}\Big(f(z), z_0\Big) = \frac{g(z_0)}{\phi(z_0)} = \frac{g(z_0)}{h'(z_0)}.$$

43. If $h(z)$ has a zero of order 2, it follows from (5) of Section 6.4 that $h(z) = (z - z_0)^2\,\phi(z)$, where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$. With $f(z) = g(z)/h(z)$, use (2) with $n = 2$ to derive a formula analogous to (4) for $\text{Res}(f(z),\ z_0)$ at a pole of order 2 at $z = z_0$.

44. Let $f(z) = 1/[p(z)]^2$ where $p(z)$ has a zero of order 1 at $z = z_0$.

(**a**) Use Problem 41 to derive the formula $\text{Res}(f(z), z_0) = -\dfrac{p''(z_0)}{[p'(z_0)]^3}$.

(**b**) Use part (b) to determine the residue of $f(z) = \dfrac{1}{(z^2 - z)^2}$ at $z = 0$.

Computer Lab Assignments

In problems 45–48, use a CAS to find the residue at each pole of the given function. In *Mathematica* use the command **Residue**. If necessary, use the command **N** to give your answer in the form $a + bi$.

45. $f(z) = \dfrac{2}{(z-1)^2(z^2+4)}$ **46.** $f(z) = \dfrac{z\cosh^2 z}{(z^2 - i)^3}$

47. $f(z) = \dfrac{e^{iz^2}\sin z}{z^2(z+i)}$ **48.** $f(z) = \dfrac{ie^{3/z}}{(z^2 + 6z + 10)^3}$

6.6 Some Consequences of the Residue Theorem

INTRODUCTION In this section we shall see how residue theory can be used to evaluate *real* integrals of the forms

$$\int_0^{2\pi} F(\cos\theta, \sin\theta)\,d\theta, \tag{1}$$

$$\int_{-\infty}^{\infty} f(x)\,dx, \tag{2}$$

$$\int_{-\infty}^{\infty} f(x)\cos\alpha x\,dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx, \tag{3}$$

where F in (1) and f in (2) and (3) are rational functions. For the rational function $f(x) = p(x)/q(x)$ in (2) and (3), we will assume that the polynomials p and q have no common factors. In addition to evaluating the three integrals just given, we shall demonstrate how to use residues to evaluate real improper integrals that require integration along a branch cut.

The discussion ends with the relationship between the residue theory and the zeros of an analytic function and a consideration of how residues can, in certain cases, be used to find the sum of an infinite series.

6.6.1 Evaluation of Real Trigonometric Integrals

Integrals of the Form $\int_0^{2\pi} F(\cos\theta, \sin\theta)\,d\theta$ The basic idea here is to convert a real trigonometric integral of form (1) into a complex integral, where the contour C is the unit circle $|z| = 1$ centered at the origin. To do this we begin with (10) of Section 2.2 to parametrize this contour by $z = e^{i\theta}$, $0 \le \theta \le 2\pi$. We can then write

$$dz = ie^{i\theta}d\theta, \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

The last two expressions follow from (2) and (3) of Section 4.3. Since $dz = ie^{i\theta}d\theta = iz\,d\theta$ and $z^{-1} = 1/z = e^{-i\theta}$, these three quantities are equivalent to

$$d\theta = \frac{dz}{iz}, \quad \cos\theta = \frac{1}{2}(z + z^{-1}), \quad \sin\theta = \frac{1}{2i}(z - z^{-1}). \tag{4}$$

The conversion of the integral in (1) into a contour integral is accomplished by replacing, in turn, $d\theta$, $\cos\theta$, and $\sin\theta$ by the expressions in (4):

$$\oint_C F\left(\frac{1}{2}(z + z^{-1}), \frac{1}{2i}(z - z^{-1})\right)\frac{dz}{iz},$$

where C is the unit circle $|z| = 1$.

EXAMPLE 1 A Real Trigonometric Integral

Evaluate $\displaystyle\int_0^{2\pi} \frac{1}{(2 + \cos\theta)^2}\,d\theta$.

Solution When we use the substitutions given in (4), the given trigonometric integral becomes the contour integral

$$\oint_C \frac{1}{\left(2 + \frac{1}{2}(z + z^{-1})\right)^2}\frac{dz}{iz} = \oint_C \frac{1}{\left(2 + \dfrac{z^2 + 1}{2z}\right)^2}\frac{dz}{iz}.$$

Carrying out the algebraic simplification of the integrand then yields

$$\frac{4}{i}\oint_C \frac{z}{(z^2 + 4z + 1)^2}\,dz.$$

From the quadratic formula we can factor the polynomial $z^2 + 4z + 1$ as $z^2 + 4z + 1 = (z - z_1)(z - z_2)$, where $z_1 = -2 - \sqrt{3}$ and $z_2 = -2 + \sqrt{3}$. Thus, the integrand can be written

$$\frac{z}{(z^2 + 4z + 1)^2} = \frac{z}{(z - z_1)^2(z - z_2)^2}.$$

Because only z_2 is inside the unit circle C, we have

$$\oint_C \frac{z}{(z^2 + 4z + 1)^2}\,dz = 2\pi i\,\text{Res}(f(z), z_2).$$

To calculate the residue, we first note that z_2 is a pole of order 2 and so we use (2) of Section 6.5:

$$\text{Res}(f(z), z_2) = \lim_{z\to z_2} \frac{d}{dz}\left[(z-z_2)^2 f(z)\right] = \lim_{z\to z_2} \frac{d}{dz}\frac{z}{(z-z_1)^2}$$
$$= \lim_{z\to z_2} \frac{-z-z_1}{(z-z_1)^3} = \frac{1}{6\sqrt{3}}.$$

Hence, $$\frac{4}{i}\oint_C \frac{z}{(z^2+4z+1)}\,dz = \frac{4}{i}\cdot 2\pi i \text{Res}(f(z), z_1) = \frac{4}{i}\cdot 2\pi i \cdot \frac{1}{6\sqrt{3}}$$

and, finally, $$\int_0^{2\pi} \frac{1}{(2+\cos\theta)^2}\,d\theta = \frac{4\pi}{3\sqrt{3}}.$$

6.6.2 Evaluation of Real Improper Integrals

Integrals of the Form $\int_{-\infty}^{\infty} f(x)\,dx$ Suppose $y = f(x)$ is a real function that is defined and continuous on the interval $[0, \infty)$. In elementary calculus the improper integral $I_1 = \int_0^{\infty} f(x)\,dx$ is defined as the limit

$$I_1 = \int_0^{\infty} f(x)\,dx = \lim_{R\to\infty}\int_0^R f(x)\,dx. \tag{5}$$

If the limit exists, the integral I_1 is said to be **convergent**; otherwise, it is **divergent**. The improper integral $I_2 = \int_{-\infty}^0 f(x)\,dx$ is defined similarly:

$$I_2 = \int_{-\infty}^0 f(x)\,dx = \lim_{R\to\infty}\int_{-R}^0 f(x)\,dx. \tag{6}$$

Finally, if f is continuous on $(-\infty, \infty)$, then $\int_{-\infty}^{\infty} f(x)\,dx$ is defined to be

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^0 f(x)\,dx + \int_0^{\infty} f(x)\,dx = I_1 + I_2, \tag{7}$$

provided *both* integrals I_1 and I_2 are convergent. If either one, I_1 or I_2, is divergent, then $\int_{-\infty}^{\infty} f(x)\,dx$ is divergent. It is important to remember that the right-hand side of (7) is *not* the same as

$$\lim_{R\to\infty}\left[\int_{-R}^0 f(x)\,dx + \int_0^R f(x)\,dx\right] = \lim_{R\to\infty}\int_{-R}^R f(x)\,dx. \tag{8}$$

For the integral $\int_{-\infty}^{\infty} f(x)\,dx$ to be convergent, the limits (5) and (6) must exist independently of one another. *But*, in the event that we know (a priori) that an improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ converges, we can then evaluate it by means of the single limiting process given in (8):

$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty}\int_{-R}^R f(x)\,dx. \tag{9}$$

On the other hand, the symmetric limit in (9) may exist even though the improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ is divergent. For example, the integral $\int_{-\infty}^{\infty} x\,dx$ is divergent since $\lim_{R\to\infty}\int_0^R x\,dx = \lim_{R\to\infty}\frac{1}{2}R^2 = \infty$. However, (9) gives

$$\lim_{R\to\infty}\int_{-R}^R x\,dx = \lim_{R\to\infty}\frac{1}{2}[R^2 - (-R)^2] = 0. \tag{10}$$

The limit in (9), if it exists, is called the **Cauchy principal value** (P.V.) of the integral and is written

$$\text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx. \tag{11}$$

In (10) we have shown that $\text{P.V.} \int_{-\infty}^{\infty} x\,dx = 0$. To summarize:

Cauchy Principal Value

When an integral of form (2) *converges, its Cauchy principal value is the same as the value of the integral. If the integral diverges, it may still possess a Cauchy principal value* (11).

Important observation about even functions ▶

One final point about the Cauchy principal value: Suppose $f(x)$ is continuous on $(-\infty, \infty)$ and is an *even* function, that is, $f(-x) = f(x)$. Then its graph is symmetric with respect to the y-axis and as a consequence

$$\int_{-R}^{0} f(x)\,dx = \int_{0}^{R} f(x)\,dx \tag{12}$$

and

$$\int_{-R}^{R} f(x)\,dx = \int_{-R}^{0} f(x)\,dx + \int_{0}^{R} f(x)\,dx = 2\int_{0}^{R} f(x)\,dx. \tag{13}$$

From (12) and (13) we conclude that *if* the Cauchy principal value (11) exists, then both $\int_0^\infty f(x)\,dx$ and $\int_{-\infty}^{\infty} f(x)\,dx$ converge. The values of the integrals are

$$\int_{0}^{\infty} f(x)\,dx = \frac{1}{2}\,\text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\,dx = \text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx.$$

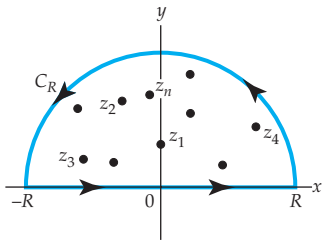


FIGURE 6.6.1 Semicircular contour

To evaluate an integral $\int_{-\infty}^{\infty} f(x)\,dx$, where the rational function $f(x) = p(x)/q(x)$ is continuous on $(-\infty, \infty)$, by residue theory we replace x by the complex variable z and integrate the complex function f over a closed contour C that consists of the interval $[-R, R]$ on the real axis and a semicircle C_R of radius large enough to enclose all the poles of $f(z) = p(z)/q(z)$ in the upper half-plane $\text{Im}(z) > 0$. See **FIGURE 6.6.1**. By Theorem 6.5.3 of Section 6.5 we have

$$\oint_C f(z)\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f(z), z_k),$$

where z_k, $k = 1, 2, \ldots, n$ denotes poles in the upper half-plane. If we can show that the integral $\int_{C_R} f(z)\,dz \to 0$ as $R \to \infty$, then we have

$$\text{P.V.} \int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx = 2\pi i \sum_{k=1}^{n} \text{Res}(f(z), z_k). \tag{14}$$

EXAMPLE 2 Cauchy P.V. of an Improper Integral

Evaluate the Cauchy principal value of $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\,dx$.

Solution Let $f(z) = 1/(z^2+1)(z^2+9)$. Since

$$(z^2+1)(z^2+9) = (z-i)(z+i)(z-3i)(z+3i),$$

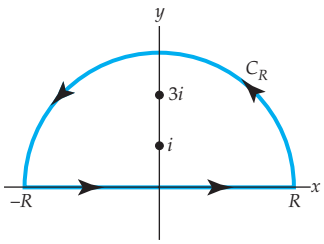


FIGURE 6.6.2 Contour for Example 2

we take C be the closed contour consisting of the interval $[-R,\ R]$ on the x-axis and the semicircle C_R of radius $R > 3$. As seen from FIGURE 6.6.2,

$$\oint_C \frac{1}{(z^2+1)(z^2+9)}\,dz = \int_{-R}^{R} \frac{1}{(x^2+1)(x^2+9)}\,dx + \int_{C_R} \frac{1}{(z^2+1)(z^2+9)}\,dz$$

$$= I_1 + I_2.$$

However, by the Residue Theorem we have:

$$I_1 + I_2 = 2\pi i\,[\text{Res}(f(z), i) + \text{Res}(f(z), 3i)].$$

At the simple poles $z = i$ and $z = 3i$ we find, respectively,

$$\text{Res}(f(z),\ i) = \frac{1}{16i} \quad \text{and} \quad \text{Res}(f(z),\ 3i) = -\frac{1}{48i},$$

so that

$$I_1 + I_2 = 2\pi i\left[\frac{1}{16i} + \left(-\frac{1}{48i}\right)\right] = \frac{\pi}{12}. \tag{15}$$

We now want to let $R \to \infty$ in (15). Before doing this, we use the inequality (8) of Section 1.2 to note that on the contour C_R,

$$\left|(z^2+1)(z^2+9)\right| = \left|z^2+1\right| \cdot \left|z^2+9\right| \geq \left|\,|z^2|\ -1\,\right| \cdot \left|\,|z^2|\ -9\,\right| = (R^2-1)(R^2-9).$$

Since the length L of the semicircle is πR, it follows from the ML-inequality, Theorem 5.2.3, that

$$|I_2| = \left|\int_{C_R} \frac{1}{(z^2+1)(z^2+9)}\,dz\right| \leq \frac{\pi R}{(R^2-1)(R^2-9)}.$$

This last result shows that $|I_2| \to 0$ as $R \to \infty$, and so we conclude that $\lim_{R\to\infty} I_2 = 0$. It follows from (15) that $\lim_{R\to\infty} I_1 = \pi/12$; in other words,

$$\lim_{R\to\infty}\int_{-R}^{R} \frac{1}{(x^2+1)(x^2+9)}\,dx = \frac{\pi}{12} \quad \text{or} \quad \text{P.V.}\int_{-\infty}^{\infty} \frac{1}{(x^2+1)(x^2+9)}\,dx = \frac{\pi}{12}.$$

Because the integrand in Example 2 is an even function, the existence of the Cauchy principal value implies that the original integral converges to $\pi/12$.

It is often tedious to have to show that the contour integral along C_R approaches zero as $R \to \infty$. Sufficient conditions under which this behavior is always true are summarized in the next theorem.

THEOREM 6.6.1 Behavior of Integral as $R \to \infty$

Suppose $f(z) = \dfrac{p(z)}{q(z)}$ is a rational function, where the degree of $p(z)$ is n and the degree of $q(z)$ is $m \geq n+2$. If C_R is a semicircular contour $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, then $\int_{C_R} f(z)\,dz \to 0$ as $R \to \infty$.

In other words, the integral along C_R approaches zero as $R \to \infty$ when the denominator of f is of a power at least 2 more than its numerator. The proof of this fact follows in the same manner as in Example 2. Notice in that example, the conditions stipulated in Theorem 6.6.1 are satisfied, since the degree of $p(z) = 1$ is 0 and the degree of $q(z) = (z^2+1)(z^2+9)$ is 4.

EXAMPLE 3 Cauchy P.V. of an Improper Integral

Evaluate the Cauchy principal value of $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^4+1}\,dx$.

Solution By inspection of the integrand we see that the conditions given in Theorem 6.6.1 are satisfied. Moreover, we know from Example 3 of Section 6.5 that $f(z) = 1/(z^4+1)$ has simple poles in the upper half-plane at $z_1 = e^{\pi i/4}$ and $z_2 = e^{3\pi i/4}$. We also saw in that example that the residues at these poles are

$$\text{Res}(f(z), z_1) = -\frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i \quad \text{and} \quad \text{Res}(f(z), z_2) = \frac{1}{4\sqrt{2}} - \frac{1}{4\sqrt{2}}i\,.$$

Thus, by (14),

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{1}{x^4+1}\,dx = 2\pi i\,[\text{Res}(f(z), z_1) + \text{Res}(f(z), z_2)] = \frac{\pi}{\sqrt{2}}.$$

Since the integrand is an even function, the original integral converges to $\pi/\sqrt{2}$. ☰

Integrals of the Form $\int_{-\infty}^{\infty} f(x)\cos\alpha x\,dx$ and $\int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx$

Because improper integrals of the form $\int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx$ are encountered in applications of Fourier analysis, they often are referred to as **Fourier integrals.** Fourier integrals appear as the real and imaginary parts in the improper integral $\int_{-\infty}^{\infty} f(x)e^{i\alpha x}\,dx$.* In view of Euler's formula $e^{i\alpha x} = \cos\alpha x + i\sin\alpha x$, where α is a positive real number, we can write

$$\int_{-\infty}^{\infty} f(x)\,e^{i\alpha x}dx = \int_{-\infty}^{\infty} f(x)\cos\alpha x\,dx + i\int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx \tag{16}$$

whenever both integrals on the right-hand side converge. Suppose $f(x) = p(x)/q(x)$ is a rational function that is continuous on $(-\infty,\ \infty)$. Then both Fourier integrals in (10) can be evaluated at the same time by considering the complex integral $\int_C f(z)e^{i\alpha z}dz$, where $\alpha > 0$, and the contour C again consists of the interval $[-R,\ R]$ on the real axis and a semicircular contour C_R with radius large enough to enclose the poles of $f(z)$ in the upper-half plane.

Before proceeding, we give, without proof, sufficient conditions under which the contour integral along C_R approaches zero as $R \to \infty$.

THEOREM 6.6.2 Behavior of Integral as $R \to \infty$

Suppose $f(z) = \dfrac{p(z)}{q(z)}$ is a rational function, where the degree of $p(z)$ is n and the degree of $q(z)$ is $m \geq n+1$. If C_R is a semicircular contour $z = Re^{i\theta}$, $0 \leq \theta \leq \pi$, and $\alpha > 0$, then $\int_{C_R} f(z)\,e^{i\alpha z}dz \to 0$ as $R \to \infty$.

EXAMPLE 4 Using Symmetry

Evaluate the Cauchy principal value of $\displaystyle\int_0^{\infty} \frac{x\sin x}{x^2+9}\,dx$.

Solution First note that the limits of integration in the given integral are not from $-\infty$ to ∞ as required by the method just described. This can be remedied by observing

*See Section 6.7.

that since the integrand is an even function of x (verify), we can write

$$\int_0^\infty \frac{x\sin x}{x^2+9}\,dx = \frac{1}{2}\int_{-\infty}^\infty \frac{x\sin x}{x^2+9}\,dx. \tag{17}$$

With $\alpha = 1$ we now form the contour integral

$$\oint_C \frac{z}{z^2+9}e^{iz}dz,$$

where C is the same contour shown in Figure 6.6.2. By Theorem 6.5.3,

$$\int_{C_R} \frac{z}{z^2+9}e^{iz}dz + \int_{-R}^R \frac{x}{x^2+9}e^{ix}\,dx = 2\pi i\ \text{Res}(f(z)e^{iz}, 3i),$$

where $f(z) = z/(z^2+9)$, and

$$\text{Res}\Big(f(z)e^{iz},\ 3i\Big) = \left.\frac{ze^{iz}}{2z}\right|_{z=3i} = \frac{e^{-3}}{2}$$

from (4) of Section 6.5. Then, from Theorem 6.6.2 we conclude $\int_{C_R} f(z)e^{iz}dz \to 0$ as $R \to \infty$, and so

$$\text{P.V.}\int_{-\infty}^\infty \frac{x}{x^2+9}e^{ix}\,dx = 2\pi i\left(\frac{e^{-3}}{2}\right) = \frac{\pi}{e^3}i.$$

But by (16),

$$\int_{-\infty}^\infty \frac{x}{x^2+9}e^{ix}\,dx = \int_{-\infty}^\infty \frac{x\cos x}{x^2+9}\,dx + i\int_{-\infty}^\infty \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{e^3}i.$$

Equating real and imaginary parts in the last line gives the bonus result

$$\text{P.V.}\int_{-\infty}^\infty \frac{x\cos x}{x^2+9}\,dx = 0 \qquad \text{along with} \qquad \text{P.V.}\int_{-\infty}^\infty \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{e^3}. \tag{18}$$

Finally, in view of the fact that the integrand is an even function, we obtain the value of the prescribed integral:

$$\int_0^\infty \frac{x\sin x}{x^2+9}\,dx = \frac{1}{2}\int_{-\infty}^\infty \frac{x\sin x}{x^2+9}\,dx = \frac{\pi}{2e^3}.$$

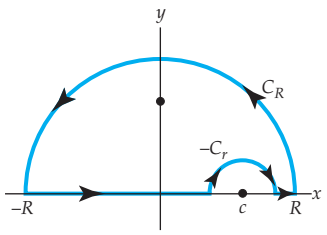


FIGURE 6.6.3 Indented contour

Indented Contours The improper integrals of forms (2) and (3) that we have considered up to this point were continuous on the interval $(-\infty, \infty)$. In other words, the complex function $f(z) = p(z)/q(z)$ did not have poles on the real axis. In the situation where f has poles on the real axis, we must modify the procedure illustrated in Examples 2–4. For example, to evaluate $\int_{-\infty}^\infty f(x)\,dx$ by residues when $f(z)$ has a pole at $z = c$, where c is a real number, we use an **indented contour** as illustrated in FIGURE 6.6.3. The symbol C_r denotes a semicircular contour centered at $z = c$ and oriented in the *positive* direction. The next theorem is important to this discussion.

THEOREM 6.6.3 Behavior of Integral as $r \to 0$

Suppose f has a simple pole $z = c$ on the real axis. If C_r is the contour defined by $z = c + re^{i\theta}$, $0 \le \theta \le \pi$, then

$$\lim_{r\to 0}\int_{C_r} f(z)\,dz = \pi i\ \text{Res}(f(z), c).$$

Proof Since f has a simple pole at $z = c$, its Laurent series is

$$f(z) = \frac{a_{-1}}{z - c} + g(z),$$

where $a_{-1} = \text{Res}(f(z),\ c)$ and g is analytic at the point c. Using the Laurent series and the parametrization of C_r we have

$$\int_{C_r} f(z)\,dz = a_{-1}\int_0^{\pi} \frac{ire^{i\theta}}{re^{i\theta}}\,d\theta + ir\int_0^{\pi} g(c + re^{i\theta})\,e^{i\theta}d\theta = I_1 + I_2. \tag{19}$$

First, we see that

$$I_1 = a_{-1}\int_0^{\pi} i\,d\theta = \pi i a_{-1} = \pi i\,\text{Res}(f(z), c).$$

Next, g is analytic at c, and so it is continuous at this point and bounded in a neighborhood of the point; that is, there exists an $M > 0$ for which $\left|g(c + re^{i\theta})\right| \le M$. Hence,

$$|I_2| = \left| ir\int_0^{\pi} g(c + re^{i\theta})\,d\theta \right| \le r\int_0^{\pi} M\,d\theta = \pi r M.$$

It follows from this last inequality that $\lim_{r\to 0} |I_2| = 0$ and consequently $\lim_{r\to 0} I_2 = 0$. By taking the limit of (19) as $r \to 0$, the theorem is proved. ☰

EXAMPLE 5 Using an Indented Contour

Evaluate the Cauchy principal value of $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 - 2x + 2)}\,dx$.

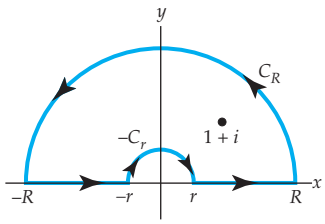


FIGURE 6.6.4 Indented contour for Example 5

Solution Since the integral is of the type given in (3), we consider the contour integral

$$\oint_C \frac{e^{iz}}{z(z^2 - 2z + 2)}\,dz.$$

The function $f(z) = 1/z(z^2 - 2z + 2)$ has a pole at $z = 0$ and at $z = 1 + i$ in the upper half-plane. The contour C, shown in FIGURE 6.6.4, is indented at the origin. Adopting an obvious condensed notation, we have

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 2\pi i\,\text{Res}(f(z)e^{iz}, 1 + i), \tag{20}$$

where $\int_{-C_r} = -\int_{C_r}$. If we take the limits of (20) as $R \to \infty$ and as $r \to 0$, it follows from Theorems 6.6.2 and 6.6.3 that

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2 - 2x + 2)}\,dx - \pi i\,\text{Res}(f(z)e^{iz}, 0) = 2\pi i\,\text{Res}(f(z)e^{iz}, 1 + i).$$

Now,

$$\text{Res}(f(z)e^{iz}, 0) = \frac{1}{2} \quad \text{and} \quad \text{Res}(f(z)e^{iz}, 1 + i) = -\frac{e^{-1+i}}{4}(1 + i).$$

Therefore,

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2 - 2x + 2)}\,dx = \pi i\left(\frac{1}{2}\right) + 2\pi i\left(-\frac{e^{-1+i}}{4}(1 + i)\right).$$

Using $e^{-1+i} = e^{-1}(\cos 1 + i \sin 1)$, simplifying, and then equating real and imaginary parts, we get from the last equality

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2 - 2x + 2)}\, dx = \frac{\pi}{2}\, e^{-1}(\sin 1 + \cos 1)$$

and

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 - 2x + 2)}\, dx = \frac{\pi}{2}[1 + e^{-1}(\sin 1 - \cos 1)].$$

6.6.3 Integration along a Branch Cut

Branch Point at $z = 0$ In the next discussion we examine integrals of the form $\int_0^{\infty} f(x)\, dx$, where the integrand $f(x)$ is algebraic. But similar to Example 5, these integrals require a special type of contour because when $f(x)$ is converted to a complex function, the resulting integrand $f(z)$ has, in addition to poles, a nonisolated singularity at $z = 0$. Before proceeding, the reader is encouraged to review the discussion on branch cuts in Sections 3.1 and 4.1.

In the example that follows we consider a special case of the real integral

$$\int_0^{\infty} \frac{x^{\alpha - 1}}{x + 1}\, dx, \tag{21}$$

where α is a real constant restricted to the interval $0 < \alpha < 1$. Observe that when $\alpha = \frac{1}{2}$ and x is replaced by z, the integrand of (12) becomes the multiple-valued function

$$\frac{1}{z^{1/2}(z + 1)}. \tag{22}$$

The origin is a branch point of (22) since $z^{1/2}$ has two values for any $z \neq 0$. If you envision traveling in a complete circle around the origin $z = 0$, starting from a point $z = re^{i\theta}$, $r > 0$, you return to the same starting point z, but θ has increased by 2π. Correspondingly, the value of $z^{1/2}$ changes from $z^{1/2} = \sqrt{r}\, e^{i\theta/2}$ to a different value or different branch:

$$z^{1/2} = \sqrt{r}\, e^{i(\theta + 2\pi)/2} = \sqrt{r}\, e^{i\theta/2} e^{i\pi} = -\sqrt{r}\, e^{i\theta/2}.$$

Recall, we can force $z^{1/2}$ to be single valued by restricting θ to some interval of length 2π. For (22), if we choose the positive x-axis as a branch cut, in other words by restricting θ to $0 < \theta < 2\pi$, we then guarantee that $z^{1/2} = \sqrt{r}\, e^{i\theta/2}$ is single valued. See page 116.

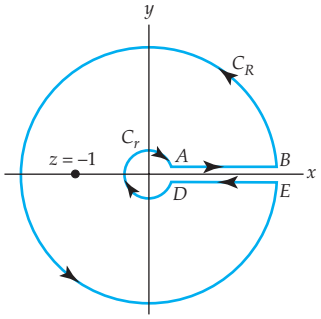


FIGURE 6.6.5 Contour for Example 6

EXAMPLE 6 Integration along a Branch Cut

Evaluate $\displaystyle\int_0^{\infty} \frac{1}{\sqrt{x}(x + 1)}\, dx.$

Solution First observe that the real integral is improper for two reasons. Notice an infinite discontinuity at $x = 0$ and the infinite limit of integration. Moreover, it can be argued from the facts that the integrand behaves like $x^{-1/2}$ near the origin and like $x^{-3/2}$ as $x \to \infty$, that the integral converges.

We form the integral $\displaystyle\oint_C \frac{1}{z^{1/2}(z + 1)}\, dz$, where C is the closed contour shown in FIGURE 6.6.5 consisting of four components: C_r and C_R are portions of circles, and

AB and *ED* are parallel horizontal line segments running along opposite sides of the branch cut. The integrand $f(z)$ of the contour integral is single valued and analytic on and within C, except for the simple pole at $z = -1 = e^{\pi i}$. Hence we can write

$$\oint_C \frac{1}{z^{1/2}(z+1)}\,dz = 2\pi i \operatorname{Res}(f(z), -1)$$

or

$$\int_{C_R} + \int_{ED} + \int_{C_r} + \int_{AB} = 2\pi i \operatorname{Res}(f(z), -1). \tag{23}$$

Despite what is shown in Figure 6.6.5, it is permissible to think that the line segments *AB* and *ED* actually rest on the positive real axis, more precisely, *AB* coincides with the *upper side* of the positive real axis for which $\theta = 0$ and *ED* coincides with the *lower side* of the positive real axis for which $\theta = 2\pi$. On *AB*, $z = xe^{0i}$, and on *ED*, $z = xe^{(0+2\pi)i} = xe^{2\pi i}$, so that

$$\int_{ED} = \int_R^r \frac{(xe^{2\pi i})^{-1/2}}{xe^{2\pi i}+1}\,(e^{2\pi i}dx) = -\int_R^r \frac{x^{-1/2}}{x+1}\,dx = \int_r^R \frac{x^{-1/2}}{x+1}\,dx \tag{24}$$

and

$$\int_{AB} = \int_r^R \frac{(xe^{0i})^{-1/2}}{xe^{0i}+1}\,(e^{0i}dx) = \int_r^R \frac{x^{-1/2}}{x+1}\,dx. \tag{25}$$

Now with $z = re^{i\theta}$ and $z = Re^{i\theta}$ on C_r and C_R, respectively, it can be shown, by analysis similar to that given in Example 2 and in the proof of Theorem 6.6.1, that $\int_{C_r} \to 0$ as $r \to 0$ and $\int_{C_R} \to 0$ as $R \to \infty$. Thus from (23), (24), and (25) we see that

$$\lim_{\substack{r\to 0\\ R\to\infty}} \left[\int_{C_R} + \int_{ED} + \int_{C_r} + \int_{AB} = 2\pi i \operatorname{Res}(f(z), -1)\right]$$

is the same as

$$2\int_0^\infty \frac{1}{\sqrt{x}(x+1)}\,dx = 2\pi i \operatorname{Res}(f(z), -1). \tag{26}$$

Finally, from (4) of Section 6.5,

$$\operatorname{Res}(f(z), -1) = z^{-1/2}\big|_{z=e^{\pi i}} = e^{-\pi i/2} = -i$$

and so (26) yields the result

$$\int_0^\infty \frac{1}{\sqrt{x}(x+1)}\,dx = \pi.$$

6.6.4 The Argument Principle and Rouché's Theorem

Argument Principle Unlike the foregoing discussion in which the focus was on the evaluation of real integrals, we next apply residue theory to the location of zeros of an analytic function. To get to that topic, we must first consider two theorems that are important in their own right.

In the first theorem we need to count the number of zeros and poles of a function f that are located within a simple closed contour C; in this counting we include the order or multiplicity of each zero and pole. For example, if

$$f(z) = \frac{(z-1)(z-9)^4(z+i)^2}{(z^2-2z+2)^2(z-i)^6(z+6i)^7} \tag{27}$$

and C is taken to be the circle $|z| = 2$, then inspection of the numerator of f reveals that the zeros inside C are $z = 1$ (a simple zero) and $z = -i$ (a zero of order or multiplicity 2). Therefore, the number N_0 of zeros inside C is taken to be

$N_0 = 1+2 = 3$. Similarly, inspection of the denominator of f shows, after factoring $z^2 - 2z + 2$, that the poles inside C are $z = 1 - i$ (pole of order 2), $z = 1 + i$ (pole of order 2), and $z = i$ (pole of order 6). The number N_p of poles inside C is taken to be $N_p = 2 + 2 + 6 = 10$.

THEOREM 6.6.4 Argument Principle

Let C be a simple closed contour lying entirely within a domain D. Suppose f is analytic in D except at a finite number of poles inside C, and that $f(z) \neq 0$ on C. Then

$$\frac{1}{2\pi i}\oint_C \frac{f'(z)}{f(z)}\,dz = N_0 - N_p, \tag{28}$$

where N_0 is the total number of zeros of f inside C and N_p is the total number of poles of f inside C. In determining N_0 and N_p, zeros and poles are counted according to their order or multiplicities.

Proof We start with a reminder that when we use the symbol $\oint_C$ for a contour, this signifies that we are integrating in the positive direction around the closed curve C.

The integrand $f'(z)/f(z)$ in (28) is analytic in and on the contour C except at the points in the interior of C where f has a zero or a pole. If z_0 is a zero of order n of f inside C, then by (5) of Section 6.4 we can write $f(z) = (z - z_0)^n \phi(z)$, where ϕ is analytic at z_0 and $\phi(z_0) \neq 0$. We differentiate f by the product rule,

$$f'(z) = (z - z_0)^n \phi'(z) + n(z - z_0)^{n-1}\phi(z),$$

and divide this expression by f. In some punctured disk centered at z_0, we have

$$\frac{f'(z)}{f(z)} = \frac{(z - z_0)^n \phi'(z) + n(z - z_0)^{n-1}\phi(z)}{(z - z_0)^n \phi(z)} = \frac{\phi'(z)}{\phi(z)} + \frac{n}{z - z_0}. \tag{29}$$

The result in (29) shows that the integrand $f'(z)/f(z)$ has a simple pole at z_0 and the residue at that point is

$$\begin{aligned} \operatorname{Res}\left(\frac{f'(z)}{f(z)},\ z_0\right) &= \lim_{z\to z_0}(z - z_0)\left[\frac{\phi'(z)}{\phi(z)} + \frac{n}{z - z_0}\right] \\ &= \lim_{z\to z_0}\left[(z - z_0)\frac{\phi'(z)}{\phi(z)} + n\right] = 0 + n = n, \end{aligned} \tag{30}$$

which is the order of the zero z_0.

Now if z_p is a pole of order m of f within C, then by (7) of Section 6.4 we can write $f(z) = g(z)/(z - z_p)^m$, where g is analytic at z_p and $g(z_p) \neq 0$. By differentiating, in this case $f(z) = (z - z_p)^{-m} g(z)$, we have

$$f'(z) = (z - z_p)^{-m} g'(z) - m(z - z_p)^{-m-1} g(z).$$

Therefore, in some punctured disk centered at z_p,

$$\frac{f'(z)}{f(z)} = \frac{(z - z_p)^{-m} g'(z) - m(z - z_p)^{-m-1} g(z)}{(z - z_p)^{-m} g(z)} = \frac{g'(z)}{g(z)} + \frac{-m}{z - z_p}. \tag{31}$$

We see from (31) that the integrand $f'(z)/f(z)$ has a simple pole at z_p. Proceeding as in (30), we also see that the residue at z_p is equal to $-m$, which is the negative of the order of the pole of f.

Finally, suppose that $z_{0_1}, z_{0_2}, \ldots, z_{0_r}$ and $z_{p_1}, z_{p_2}, \ldots, z_{p_s}$ are the zeros and poles of f within C and suppose further that the order of the zeros are $n_1, n_2, \ldots, n_r$ and that order of the poles are $m_1, m_2, \ldots, m_s$. Then each of these points is a simple pole of the integrand $f'(z)/f(z)$ with corresponding residues $n_1, n_2, \ldots, n_r$ and

$-m_1, -m_2, \ldots, -m_s$. It follows from the residue theorem (Theorem 6.5.3) that $\oint_c [f'(z)/f(z)]\,dz$ is equal to $2\pi i$ times the sum of the residues at the poles:

$$\oint_C \frac{f'(z)}{f(z)}\,dz = 2\pi i\left[\sum_{k=1}^{r} \text{Res}\left(\frac{f'(z)}{f(z)},\ z_{0_k}\right) + \sum_{k=1}^{s} \text{Res}\left(\frac{f'(z)}{f(z)},\ z_{p_k}\right)\right]$$
$$= 2\pi i\left[\sum_{k=1}^{r} n_k + \sum_{k=1}^{s} (-m_k)\right] = 2\pi i[N_0 - N_p].$$

Dividing by $2\pi i$ establishes (28). ☰

To illustrate Theorem 6.6.4, suppose the simple closed contour is $|z| = 2$ and the function f is the one given in (27). The result in (28) indicates that in the evaluation of $\oint_c [f'(z)/f(z)]\,dz$, each zero of f within C contributes $2\pi i$ times the order of multiplicity of the zero and each pole contributes $2\pi i$ times the negative of the order of the pole:

$$\oint_C \frac{f'(z)}{f(z)}\,dz = \overbrace{[2\pi i(1) + 2\pi i(2)]}^{\substack{\text{contribution}\\ \text{of zeros of } f}} + \overbrace{[2\pi i(-2) + 2\pi i(-2) + 2\pi i(-6)]}^{\substack{\text{contribution}\\ \text{of poles of } f}} = -14\pi i.$$

Why the Name? Why is Theorem 6.6.4 called *the argument principle*? This question may have occurred to you since no reference is made in the proof of the theorem to any arguments of complex quantities. But in point of fact there is a relation between the number $N_0 - N_p$ in Theorem 6.6.4 and $\arg(f(z))$. More precisely,

$$N_0 - N_p = \frac{1}{2\pi}\,[\text{change in arg } (f(z)) \text{ as } z \text{ traverses } C \text{ once in the positive direction}].$$

This principle can be easily verified using the simple function $f(z) = z^2$ and the unit circle $|z| = 1$ as the simple closed contour C in the z-plane. Because the function f has a zero of multiplicity 2 within C and no poles, we have $N_0 - N_p = 2$. Now, if C is parametrized by $z = e^{i\theta}$, $0 \le \theta \le 2\pi$, then its image C' in the w-plane under the mapping $w = z^2$ is $w = e^{i2\theta}$, $0 \le \theta \le 2\pi$, which is the unit circle $|w| = 1$. As z traverses C once starting at $z = 1$ ($\theta = 0$) and finishing at $z = 1$ ($\theta = 2\pi$), we see $\arg(f(z)) = \arg(w) = 2\theta$ increases from 0 to 4π. Put another way, w traverses or *winds* around the circle $|w| = 1$ twice. Thus,

$$\frac{1}{2\pi}[\text{change in arg}(f(z)) \text{ as } z \text{ traverses } C \text{ once in the positive direction}] = \frac{1}{2\pi}[4\pi - 0] = 2.$$

Rouché's Theorem The next result follows as a consequence of the argument principle. The theorem is helpful in determining the number of zeros of an analytic function.

THEOREM 6.6.5 Rouché's Theorem

Let C be a simple closed contour lying entirely within a domain D. Suppose f and g are analytic in D. If the strict inequality $|f(z) - g(z)| < |f(z)|$ holds for all z on C, then f and g have the same number of zeros (counted according to their order or multiplicities) inside C.

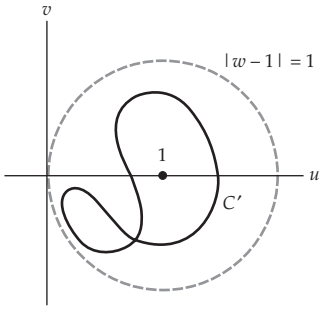


FIGURE 6.6.6 Image of C lies within the disk $|w-1|<1$.

Proof We start with the observation that the hypothesis "the strict inequality $|f(z)-g(z)|<|f(z)|$ holds for all z on C" indicates that both f and g have no zeros on the contour C. From $|f(z)-g(z)|=|g(z)-f(z)|$, we see that by dividing the inequality by $|f(z)|$ we have, for all z on C,

$$|F(z)-1|<1, \tag{32}$$

where $F(z)=g(z)/f(z)$. The inequality in (32) shows that the image C' in the w-plane of the curve C under the mapping $w=F(z)$ is a closed path and must lie within the unit open disk $|w-1|<1$ centered at $w=1$. See FIGURE 6.6.6. As a consequence, the curve C' does not enclose $w=0$, and therefore $1/w$ is analytic in and on C'. By the Cauchy-Goursat theorem,

$$\int_{C'}\frac{1}{w}\,dw=0 \qquad \text{or} \qquad \oint_C \frac{F'(z)}{F(z)}\,dz=0, \tag{33}$$

since $w=F(z)$ and $dw=F'(z)\,dz$. From the quotient rule,

$$F'(z)=\frac{f(z)g'(z)-g(z)f'(z)}{[f(z)]^2},$$

we get

$$\frac{F'(z)}{F(z)}=\frac{g'(z)}{g(z)}-\frac{f'(z)}{f(z)}.$$

Using the last expression in the second integral in (33) then gives

$$\oint_C\left[\frac{g'(z)}{g(z)}-\frac{f'(z)}{f(z)}\right]dz=0 \qquad \text{or} \qquad \oint_C\frac{g'(z)}{g(z)}\,dz=\oint_C\frac{f'(z)}{f(z)}\,dz.$$

It follows from (28) of Theorem 6.6.4, with $N_p=0$, that the number of zeros of g inside C is the same as the number of zeros of f inside C. ☰

EXAMPLE 7 Location of Zeros

Locate the zeros of the polynomial function $g(z)=z^9-8z^2+5$.

Solution We begin by choosing $f(z)=z^9$ because it has the same number of zeros as g. Since f has a zero of order 9 at the origin $z=0$, we begin our search for the zeros of g by examining circles centered at $z=0$. In other words, *if* we can establish that $|f(z)-g(z)|<|f(z)|$ for all z on *some* circle $|z|=R$, then Theorem 6.6.5 states that f and g have the same number of zeros within the disk $|z|<R$. Now by the triangle inequality (9) of Section 1.2,

$$|f(z)-g(z)|=\left|z^9-(z^9-8z^2+5)\right|=\left|8z^2-5\right|\le 8|z|^2+5.$$

Also, $|f(z)|=|z|^9$. Observe that $|f(z)-g(z)|<|f(z)|$ or $8|z|^2+5<|z|^9$ is *not* true for all points on the circle $|z|=1$, so we can draw no conclusion. However, by expanding the search to the larger circle $|z|=\frac{3}{2}$ we see

$$|f(z)-g(z)|\le 8|z|^2+5=8\left(\tfrac{3}{2}\right)^2+5=23<\left(\tfrac{3}{2}\right)^9=|f(z)| \tag{34}$$

since $\left(\frac{3}{2}\right)^9\approx 38.44$. We conclude from (34) that because f has a zero of order 9 within the disk $|z|<\frac{3}{2}$, all nine zeros of g lie within the same disk. ☰

By slightly subtler reasoning, we can demonstrate that the function g in Example 7 has *some* zeros inside the unit disk $|z| < 1$. To see this suppose we choose $f(z) = -8z^2 + 5$. Then for all z on $|z| = 1$,

$$|f(z) - g(z)| = \left|(-8z^2 + 5) - (z^9 - 8z^2 + 5)\right| = \left|-z^9\right| = |z|^9 = (1)^9 = 1. \tag{35}$$

But from (10) of Section 1.2 we have, for all z on $|z| = 1$,

$$|f(z)| = |-f(z)| = \left|8z^2 - 5\right| \geq \left|8|z|^2 - |5|\right| = |8 - 5| = 3. \tag{36}$$

The values in (35) and (36) show, for all z on $|z| = 1$, that $|f(z) - g(z)| < |f(z)|$. Because f has two zeros within $|z| < 1$ (namely, $\pm\sqrt{\frac{5}{8}} \approx \pm 0.79$), we can conclude from Theorem 6.6.5 that two zeros of g also lie within this disk.

We can continue the reasoning of the previous paragraph. Suppose now we choose $f(z) = 5$ and $|z| = \frac{1}{2}$. Then for all z on $|z| = \frac{1}{2}$,

$$|f(z) - g(z)| = \left|5 - (z^9 - 8z^2 + 5)\right| = \left|-z^9 + 8z^2\right| \leq |z|^9 + 8|z|^2 = \left(\tfrac{1}{2}\right)^9 + 2 \approx 2.002.$$

We now have $|f(z) - g(z)| < |f(z)| = 5$ for all z on $|z| = \frac{1}{2}$. Since f has no zeros within the disk $|z| < \frac{1}{2}$, neither does g. At this point we are able to conclude that all nine zeros of $g(z) = z^9 - 8z^2 + 5$ lie within the annular region $\frac{1}{2} < |z| < \frac{3}{2}$; two of these zeros lie within $\frac{1}{2} < |z| < 1$.

6.6.5 Summing Infinite Series

Using cot πz In some specialized circumstances, the residues at the simple poles of the trigonometric function $\cot \pi z$ enable us to find the sum of an infinite series.

In Section 4.3 we saw that the zeros of $\sin z$ were the real numbers $z = k\pi$, $k = 0, \pm 1, \pm 2, \ldots$. Thus the function $\cot \pi z$ has simple poles at the zeros of $\sin \pi z$, which are $\pi z = k\pi$ or $z = k$, $k = 0, \pm 1, \pm 2, \ldots$. If a polynomial function $p(z)$ has (*i*) real coefficients, (*ii*) degree $n \geq 2$, and (*iii*) no *integer* zeros, then the function

$$f(z) = \frac{\pi \cot \pi z}{p(z)} \tag{37}$$

has an infinite number of simple poles $z = 0, \pm 1, \pm 2, \ldots$ from $\cot \pi z$ and a finite number of poles $z_{p_1}, z_{p_2}, \ldots, z_{p_r}$ from the zeros of $p(z)$. The closed rectangular contour C shown in FIGURE 6.6.7 has vertices $\left(n + \frac{1}{2}\right) + ni$, $-\left(n + \frac{1}{2}\right) + ni$, $-\left(n + \frac{1}{2}\right) - ni$, and $\left(n + \frac{1}{2}\right) - ni$, where n is taken large enough so that C encloses the simple poles $z = 0, \pm 1, \pm 2, \ldots, \pm n$ and *all* of the poles $z_{p_1}, z_{p_2}, \ldots, z_{p_r}$. By the residue theorem,

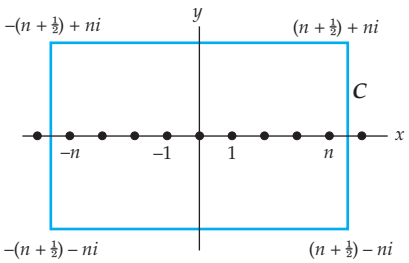


FIGURE 6.6.7 Rectangular contour C enclosing poles of (37)

$$\oint_C \frac{\pi \cot \pi z}{p(z)}\, dz = 2\pi i \left[\sum_{k=-n}^{n} \text{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ k\right) + \sum_{j=1}^{r} \text{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ z_{p_j}\right) \right]. \tag{38}$$

In a manner similar to that used several times in the discussion in Subsection 6.6.2, it can be shown that $\oint_C \pi \cot \pi z\, dz/p(z) \to 0$ as $n \to \infty$ and so (38) becomes $0 = \sum_k \text{residues} + \sum_j \text{residues}$. That is,

$$\sum_{k=-\infty}^{\infty} \text{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ k\right) = -\sum_{j=1}^{r} \text{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ z_{p_j}\right). \tag{39}$$

Now from (4) of Section 6.5 (with the identifications $g(z) = \pi \cos \pi z/p(z)$, $h(z) = \sin \pi z$, $h'(z) = \pi \cos \pi z$), it is a straightforward task to compute the residues at the simple poles $0, \pm 1, \pm 2, \ldots$:

$$\operatorname{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ k\right) = \frac{\pi \cos k\pi/p(k)}{\pi \cos k\pi} = \frac{1}{p(k)}. \tag{40}$$

By combining (40) and (39) we arrive at our desired result

$$\sum_{k=-\infty}^{\infty} \frac{1}{p(k)} = -\sum_{j=1}^{r} \operatorname{Res}\left(\frac{\pi \cot \pi z}{p(z)},\ z_{p_j}\right). \tag{41}$$

Using csc πz There exist several more summation formulas similar to (41). If $p(z)$ is a polynomial function satisfying the same assumptions (i)–(iii) given above, then the function

$$f(z) = \frac{\pi \csc \pi z}{p(z)} \tag{42}$$

has an infinite number of simple poles $z = 0, \pm 1, \pm 2, \ldots$ from $\csc \pi z$ and a finite number of poles $z_{p_1}, z_{p_2}, \ldots, z_{p_r}$ from the zeros of $p(z)$. In this case it can be shown that

$$\sum_{k=-\infty}^{\infty} \frac{(-1)^k}{p(k)} = -\sum_{j=1}^{r} \operatorname{Res}\left(\frac{\pi \csc \pi z}{p(z)},\ z_{p_j}\right). \tag{43}$$

In our last example we show how to use the result in (41) to find the sum of an infinite series.

EXAMPLE 8 Summing an Infinite Series

Find the sum of the series $\displaystyle\sum_{k=0}^{\infty} \frac{1}{k^2+4}$.

Solution Observe that if we identify $p(z) = z^2 + 4$, then the three assumptions (i)–(iii) preceding (37) hold true. The zeros of $p(z)$ are $\pm 2i$ and correspond to simple poles of $f(z) = \pi \cot \pi z/(z^2+4)$. According to the formula in (41),

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2+4} = -\left[\operatorname{Res}\left(\frac{\pi \cot \pi z}{z^2+4}, -2i\right) + \operatorname{Res}\left(\frac{\pi \cot \pi z}{z^2+4}, 2i\right)\right]. \tag{44}$$

Now again by (4) of Section 6.5 we have

$$\operatorname{Res}\left(\frac{\pi \cot \pi z}{z^2+4}, -2i\right) = \frac{\pi \cot 2\pi i}{4i} \quad \text{and} \quad \operatorname{Res}\left(\frac{\pi \cot \pi z}{z^2+4}, 2i\right) = \frac{\pi \cot 2\pi i}{4i}.$$

The sum of the residues is $(\pi/2i)\cot 2\pi i$. This sum is a real quantity because from (27) of Section 4.3:

$$\frac{\pi}{2i}\cot 2\pi i = \frac{\pi}{2i}\frac{\cosh(-2\pi)}{(-i\sinh(-2\pi))} = -\frac{\pi}{2}\coth 2\pi.$$

Hence (44) becomes

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2+4} = \frac{\pi}{2}\coth 2\pi. \tag{45}$$

This is not quite the desired result. To that end we must manipulate the summation $\sum_{k=-\infty}^{\infty}$ in order to put it in the form $\sum_{k=0}^{\infty}$. Observe

$$\sum_{k=-\infty}^{\infty}\frac{1}{k^2+4}=\sum_{k=-\infty}^{-1}\frac{1}{k^2+4}+\overbrace{\frac{1}{4}}^{\substack{k=0\\ \text{term}}}+\sum_{k=1}^{\infty}\frac{1}{k^2+4}$$

$$=\sum_{k=1}^{\infty}\frac{1}{(-k)^2+4}+\frac{1}{4}+\sum_{k=1}^{\infty}\frac{1}{k^2+4}$$

$$=2\sum_{k=1}^{\infty}\frac{1}{k^2+4}+\frac{1}{4}\ =2\sum_{k=0}^{\infty}\frac{1}{k^2+4}-\frac{1}{4}. \tag{46}$$

Finally, we obtain the sum of the original series by combining (45) with (46),

$$\sum_{k=-\infty}^{\infty}\frac{1}{k^2+4}=2\sum_{k=0}^{\infty}\frac{1}{k^2+4}-\frac{1}{4}=\frac{\pi}{2}\coth 2\pi,$$

and solving for $\sum_{k=0}^{\infty}$:

$$\sum_{k=0}^{\infty}\frac{1}{k^2+4}=\frac{1}{8}+\frac{\pi}{4}\coth 2\pi. \tag{47}$$

With the help of a calculator, we find that the right side of (47) is approximately 0.9104. ☰

EXERCISES 6.6

Answers to selected odd-numbered problems begin on page ANS-18.

6.6.1 Evaluation of Real Trigonometric Integrals

In Problems 1–12, evaluate the given trigonometric integral.

1. $\displaystyle\int_0^{2\pi}\frac{1}{1+0.5\sin\theta}\,d\theta$

2. $\displaystyle\int_0^{2\pi}\frac{1}{10-6\cos\theta}\,d\theta$

3. $\displaystyle\int_0^{2\pi}\frac{\cos\theta}{3+\sin\theta}\,d\theta$

4. $\displaystyle\int_0^{2\pi}\frac{1}{1+3\cos^2\theta}\,d\theta$

5. $\displaystyle\int_0^{\pi}\frac{1}{2-\cos\theta}\,d\theta$ [*Hint*: Let $t=2\pi-\theta$.]

6. $\displaystyle\int_0^{\pi}\frac{1}{1+\sin^2\theta}d\theta$

7. $\displaystyle\int_0^{2\pi}\frac{\sin^2\theta}{5+4\cos\theta}d\theta$

8. $\displaystyle\int_0^{2\pi}\frac{\cos^2\theta}{3-\sin\theta}d\theta$

9. $\displaystyle\int_0^{2\pi}\frac{\cos 2\theta}{5-4\cos\theta}d\theta$

10. $\displaystyle\int_0^{2\pi}\frac{1}{\cos\theta+2\sin\theta+3}d\theta$

11. $\displaystyle\int_0^{2\pi}\frac{\cos^2\theta}{2+\sin\theta}d\theta$

12. $\displaystyle\int_0^{2\pi}\frac{\cos 3\theta}{5-4\cos\theta}d\theta$

In Problems 13 and 14, establish the given general result. Use Problem 13 to verify the answer in Example 1. Use Problem 14 to verify the answer to Problem 7.

13. $\displaystyle\int_0^{\pi}\frac{d\theta}{(a+\cos\theta)^2}d\theta=\frac{a\pi}{(\sqrt{a^2-1})^3},\ a>1$

14. $\displaystyle\int_0^{2\pi}\frac{\sin^2\theta}{a+b\cos\theta}d\theta=\frac{2\pi}{b^2}\left(a-\sqrt{a^2-b^2}\right),\ a>b>0$

6.6.2 Evaluation of Real Improper Integrals

In Problems 15–26, evaluate the Cauchy principal value of the given improper integral.

15. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2 - 2x + 2}\,dx$

16. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2 - 6x + 25}\,dx$

17. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^2 + 4)^2}\,dx$

18. $\displaystyle\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 1)^2}\,dx$

19. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^2 + 1)^3}\,dx$

20. $\displaystyle\int_{-\infty}^{\infty} \frac{x}{(x^2 + 4)^3}\,dx$

21. $\displaystyle\int_{-\infty}^{\infty} \frac{2x^2 - 1}{x^4 + 5x^2 + 4}\,dx$

22. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^2 + 1)^2(x^2 + 9)}\,dx$

23. $\displaystyle\int_{0}^{\infty} \frac{x^2 + 1}{x^4 + 1}\,dx$

24. $\displaystyle\int_{0}^{\infty} \frac{1}{x^6 + 1}\,dx$

25. $\displaystyle\int_{0}^{\infty} \frac{x^2}{x^6 + 1}\,dx$

26. $\displaystyle\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 2x + 2)(x^2 + 1)^2}\,dx$

In Problems 27–38, evaluate the Cauchy principal value of the given improper integral.

27. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1}\,dx$

28. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos\ 2x}{x^2 + 1}\,dx$

29. $\displaystyle\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 1}\,dx$

30. $\displaystyle\int_{0}^{\infty} \frac{\cos x}{(x^2 + 4)^2}\,dx$

31. $\displaystyle\int_{0}^{\infty} \frac{\cos\ 3x}{(x^2 + 1)^2}\,dx$

32. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x^2 + 4x + 5}\,dx$

33. $\displaystyle\int_{0}^{\infty} \frac{\cos\ 2x}{x^4 + 1}\,dx$

34. $\displaystyle\int_{0}^{\infty} \frac{x \sin x}{x^4 + 1}\,dx$

35. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + 1)(x^2 + 9)}\,dx$

36. $\displaystyle\int_{0}^{\infty} \frac{x \sin x}{(x^2 + 1)(x^2 + 4)}\,dx$

37. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x + i}\,dx$ [*Hint*: First substitute $\sin x = (e^{ix} - e^{-ix})/2i$, x real.]

38. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x + x \sin x}{x^2 + 1}\,dx$ [*Hint*: Consider $e^{iz}/(z - i)$.]

In Problems 39–42, use an indented contour and residues to establish the Cauchy principal value of the given improper integral.

39. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi$

40. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 + 1)}\,dx = \pi(1 - e^{-1})$

41. $\displaystyle\int_{0}^{\infty} \frac{1 - \cos x}{x^2}\,dx = \frac{\pi}{2}$

42. $\displaystyle\int_{-\infty}^{\infty} \frac{x \cos x}{x^2 - 3x + 2}\,dx = \pi[\sin 1 - 2 \sin 2]$

6.6.3 Integration along a Branch Cut

In Problems 43–46, proceed as in Example 6 to establish the Cauchy principal value for the given improper integral.

43. $\displaystyle\int_{0}^{\infty} \frac{1}{\sqrt{x}(x^2 + 1)}\,dx = \frac{\pi}{\sqrt{2}}$

44. $\displaystyle\int_{0}^{\infty} \frac{1}{\sqrt{x}(x + 1)(x + 4)}\,dx = \frac{\pi}{3}$

45. $\displaystyle\int_{0}^{\infty} \frac{\sqrt{x}}{(x^2 + 1)^2}\,dx = \frac{\pi}{4\sqrt{2}}$

46. $\displaystyle\int_{0}^{\infty} \frac{x^{1/3}}{(x + 1)^2}\,dx = \frac{2\pi}{3\sqrt{3}}$

In Problems 47 and 48, establish the Cauchy principal value for the given improper integral. Use Problem 47 to verify the answer in Example 6. Use Problem 48 to verify the answer to Problem 45.

47. $\displaystyle\int_0^\infty \frac{x^{\alpha-1}}{x+1}\,dx = \frac{\pi}{\sin\alpha\pi}, 0 < \alpha < 1,$

48. $\displaystyle\int_0^\infty \frac{x^{\alpha}}{(x^2+1)^2}\,dx = \frac{\pi(1-\alpha)}{4\cos(\alpha\pi/2)}, -1 < \alpha < 3,\ \alpha \neq 1$

Miscellaneous Real Integrals

49. Use the contour C shown in **FIGURE 6.6.8** to show that

$$\text{P.V.}\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x}dx = \frac{\pi}{\sin\alpha\pi},\ 0 < \alpha < 1.$$

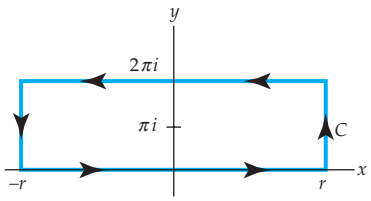


FIGURE 6.6.8 Figure for Problem 49

50. The integral result $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$ can be established using elementary calculus and polar coordinates. Use this result, the contour integral $\oint_C e^{-z^2}e^{i\alpha z}dz$, and the contour C shown in **FIGURE 6.6.9**, to show that

$$\text{P.V.}\int_0^{\infty} e^{-x^2}\cos\alpha x\,dx = \frac{\sqrt{\pi}}{2}e^{-\alpha^2/4}.$$

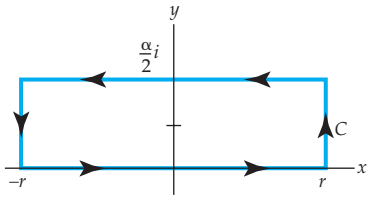


FIGURE 6.6.9 Figure for Problem 50

51. Discuss how to evaluate the Cauchy principal value of

$$\int_0^{\infty} \frac{x^{\alpha-1}}{x-1}\,dx,\ 0 < \alpha < 1.$$

Carry out your ideas.

52. (**a**) Use a graphics calculator or computer graphing program to plot on the same coordinates axes the graphs of $\sin\theta$ and $2\theta/\pi$ on the interval $0 \le \theta \le \pi/2$. Explain in graphical terms the validity of the inequality $\sin\theta \ge 2\theta/\pi$ on the interval $0 \le \theta \le \pi/2$. Use this inequality to prove that for $R > 0$,

$$\int_0^{\pi/2} e^{-R\sin\theta}d\theta < \frac{\pi}{2R}.$$

(**b**) Explain how the result in part (a) leads us to conclude that for $R > 0$,

$$\int_0^{\pi} e^{-R\sin\theta}d\theta < \frac{\pi}{R}. \tag{48}$$

The result in (48) is known as **Jordan's inequality,** which is often useful when evaluating integrals of the form $\int_{-\infty}^{\infty} f(x)\cos\alpha x\,dx$ and $\int_{-\infty}^{\infty} f(x)\sin\alpha x\,dx$.

53. Reconsider the integral in Problem 39 along with the indented contour in Figure 6.6.4. Use Jordan's inequality in Problem 52 to show that $\int_{C_R} \to 0$ as $R \to \infty$.

54. Investigate the integral $\displaystyle\int_0^{2\pi} \frac{1}{a - \sin\theta}\, d\theta$, $|a| \leq 1$, in light of the evaluation procedure outlined in Subsection 6.6.1.

55. Use Euler's formula as a starting point in the evaluation of the integral

$$\int_0^{2\pi} e^{\cos\theta}[\cos(\sin\theta - n\theta) + i\sin(\sin\theta - n\theta)]\, d\theta, n = 0, 1, 2, \ldots .$$

56. From your work in Problem 55, discern the values of the real integrals

$$\int_0^{2\pi} e^{\cos\theta}\cos(\sin\theta - n\theta)\, d\theta \quad \text{and} \quad \int_0^{2\pi} e^{\cos\theta}\sin(\sin\theta - n\theta)\, d\theta.$$

57. Suppose a real function f is continuous on the interval $[a, b]$ except at a point c within the interval. Then the **principal value** of the integral is defined by

$$\text{P.V.}\int_a^b f(x)\, dx = \lim_{\varepsilon\to 0}\left[\int_a^{c-\varepsilon} f(x)\, dx + \int_{c+\varepsilon}^b f(x)\, dx\right],\ \varepsilon > 0.$$

Compute the principal value of $\displaystyle\int_0^3 \frac{1}{x-1}\, dx$.

58. Determine whether the integral in Problem 57 converges.

6.6.4 The Argument Principle and Rouché's Theorem

In Problems 59 and 60, use the argument principle in (28) of Theorem 6.6.4 to evaluate the integral $\displaystyle\oint_C \frac{f'(z)}{f(z)}\, dz$ for the given function f and closed contour C.

59. $f(z) = z^6 - 2iz^4 + (5 - i)z^2 + 10$, C encloses all the zeros of f

60. $f(z) = \dfrac{(z - 3iz - 2)^2}{z(z^2 - 2z + 2)^5}$, C is $|z| = \frac{3}{2}$

In Problems 61–64, use the argument principle in (28) of Theorem 6.6.4 to evaluate the given integral on the indicated closed contour C. You will have to identify $f(z)$ and $f'(z)$.

61. $\displaystyle\oint_C \frac{2z+1}{z^2+z}\, dz$, C is $|z| = 2$

62. $\displaystyle\oint_C \frac{z}{z^2+4}\, dz$, C is $|z| = 3$

63. $\displaystyle\oint_C \cot z\, dz$, C is the rectangular contour with vertices $10 + i$, $-4 + i$, $-4 - i$, and $10 - i$.

64. $\displaystyle\oint_C \tan \pi z\, dz$, C is $|z - 1| = 2$

65. Use Rouché's theorem (Theorem 6.6.5) to show that all seven of the zeros of $g(z) = z^7 + 10z^3 + 14$ lie within the annular region $1 < |z| < 2$.

66. (**a**) Use Rouché's theorem (Theorem 6.6.5) to show that all four of the zeros of $g(z) = 4z^4 + 2(1 - i)z + 1$ lie within the disk $|z| < 1$.

(**b**) Show that three of the zeros of the function g in part (a) lie within the annular region $\frac{1}{2} < |z| < 1$.

67. In the proof of Theorem 6.6.5, explain how the hypothesis that the strict inequality $|f(z) - g(z)| < |f(z)|$ holds for all z on C implies that f and g cannot have zeros on C.